

TITANIC SURVIVAL PREDICTION

A Machine Learning Project Report

Submitted by

Afraz Hussain

Internship Program

eduexpose

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1. Abstract

The sinking of the RMS Titanic in 1912 remains one of the most well-documented maritime disasters in history. This project applies machine learning techniques to predict whether a given passenger would have survived the disaster, based on attributes such as age, sex, passenger class, fare paid, and number of family members aboard.

Using the Titanic dataset published on Kaggle, this project follows a complete data science workflow: data cleaning and preprocessing, exploratory data analysis (EDA) to uncover survival patterns, feature engineering, and the training of a Random Forest Classifier.

The final model achieved a prediction accuracy of 82.12% on the validation set. Analysis of feature importance revealed that Sex, Fare, and Age were the most influential predictors of survival, consistent with the well-known "women and children first" evacuation protocol followed during the disaster.

This report documents each stage of the project — from raw data to a working predictive model — and discusses the insights gained, along with directions for future improvement.

2. Introduction

Machine learning has become an essential tool for extracting patterns and making predictions from historical data. Classification problems, in particular, are widely used in real-world applications ranging from medical diagnosis to fraud detection.

The Titanic Survival Prediction problem is a classic introductory project in the data science community, made popular by Kaggle's "Titanic: Machine Learning from Disaster" competition. It provides a well-structured, real-world dataset that is ideal for learning the end-to-end machine learning pipeline.

On April 15, 1912, the RMS Titanic sank after colliding with an iceberg, resulting in the deaths of over 1,500 of the 2,224 passengers and crew aboard. While survival involved an element of chance, certain groups — such as women, children, and first-class passengers — had a significantly higher chance of survival due to evacuation protocols and access to lifeboats.

This project explores those patterns computationally, using a supervised classification approach to predict survival outcomes from passenger data.

3. Objectives

- To explore and understand the Titanic dataset through statistical summaries and visualizations.
- To clean and preprocess the data by handling missing values and encoding categorical variables.
- To identify key factors that influenced passenger survival through exploratory data analysis.
- To build a machine learning classification model capable of predicting passenger survival.
- To evaluate model performance using appropriate accuracy metrics.
- To determine feature importance and interpret the model's decision-making process.
- To generate survival predictions for the unseen test dataset and prepare a submission file.

4. Dataset Description

The dataset used in this project is the Titanic dataset, sourced from Kaggle's "Titanic: Machine Learning from Disaster" competition. It consists of two files: train.csv (used for model training) and test.csv (used for generating final predictions).

Key Features

Column	Description
PassengerId	Unique identifier for each passenger
Survived	Target variable (0 = Did not survive, 1 = Survived)
Pclass	Passenger ticket class (1 = First, 2 = Second, 3 = Third)
Sex	Gender of the passenger
Age	Age of the passenger in years
SibSp	Number of siblings/spouses aboard
Parch	Number of parents/children aboard
Fare	Passenger fare paid
Cabin	Cabin number (largely missing)
Embarked	Port of embarkation (C, Q, or S)

Dataset Size

- Training set: passenger records with known survival outcomes.
- Test set: passenger records with survival outcomes withheld, used for final prediction.

5. Data Preprocessing

Real-world datasets are rarely clean. The Titanic dataset contains several missing values and categorical fields that require preprocessing before they can be used in a machine learning model.

5.1 Handling Missing Values

- Age: Missing values filled using the median age, which is robust to outliers and skewed distributions.
- Embarked: Missing values filled using the mode (most frequently occurring port).
- Cabin: A large proportion of values were missing; these were filled with the placeholder "Unknown" rather than dropping the column entirely.
- Fare (test set only): Missing values filled using the median fare.

5.2 Encoding Categorical Variables

Machine learning models require numerical input. The categorical columns Sex and Embarked were transformed into numerical form using Label Encoding (via scikit-learn's LabelEncoder), which assigns a unique integer to each category.

5.3 Feature Selection

The following seven features were selected as model inputs based on their relevance to survival outcome: Pclass, Sex, Age, SibSp, Parch, Fare, and Embarked.

6. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand the underlying patterns in the data and identify relationships between passenger attributes and survival outcomes.

6.1 Survival Count

The overall distribution of survival outcomes shows a class imbalance, with significantly more passengers who did not survive than those who did.

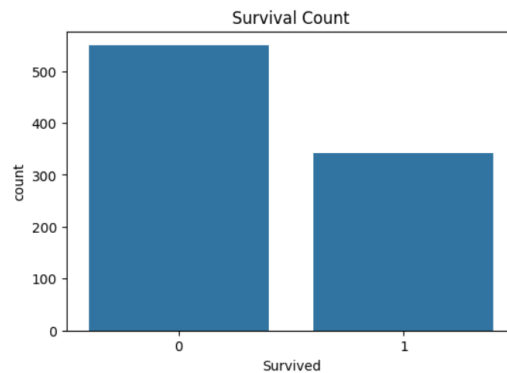


Figure 6.1 — Survival Count (0 = Did Not Survive, 1 = Survived)

6.2 Gender Distribution

The dataset contains a higher proportion of male passengers compared to female passengers.

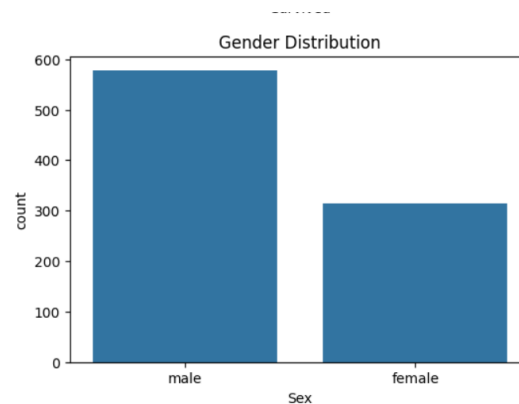


Figure 6.2 — Gender Distribution of Passengers

6.3 Survival by Gender

This is one of the most revealing visualizations in the dataset. Female passengers had a dramatically higher survival rate compared to male passengers, reflecting the “women and children first” evacuation policy.

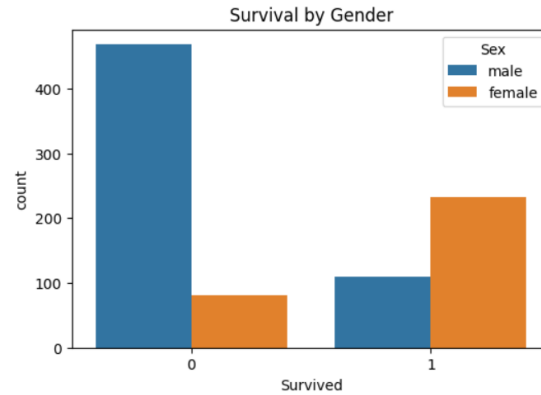


Figure 6.3 — Survival Outcome Grouped by Gender

6.4 Age Distribution

The age distribution is right-skewed, with the majority of passengers between 20 and 40 years old, and a notable concentration of passengers in their late twenties.

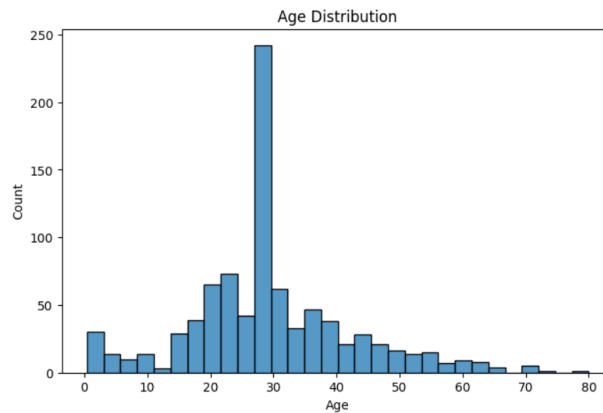


Figure 6.4 — Age Distribution of Passengers

6.5 Passenger Class Distribution

The majority of passengers traveled in third class, followed by first class and second class. This is relevant since passenger class showed a strong relationship with survival likelihood.

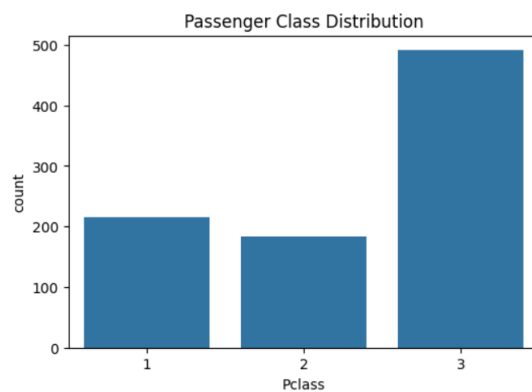


Figure 6.5 — Distribution of Passengers Across Ticket Classes

7. Model Building

After preprocessing and feature selection, the dataset was split into training and validation subsets, and a classification model was trained to predict passenger survival.

7.1 Train-Test Split

The training data was split into 80% for model training and 20% for validation, using scikit-learn's `train_test_split` function with a fixed `random_state` of 42 to ensure reproducibility.

7.2 Algorithm Selection

A Random Forest Classifier was selected for this task. Random Forest is an ensemble learning method that constructs multiple decision trees during training and outputs the mode of their individual predictions, making it robust to overfitting and well-suited to mixed categorical/numerical feature sets like this one.

Model configuration: `n_estimators = 100`, `random_state = 42`.

7.3 Model Workflow

- Load and clean the Titanic dataset (train.csv, test.csv).
- Handle missing values in Age, Embarked, Cabin, and Fare.
- Encode categorical features (Sex, Embarked) using Label Encoding.
- Select relevant features: Pclass, Sex, Age, SibSp, Parch, Fare, Embarked.
- Split data into training (80%) and validation (20%) sets.
- Train a Random Forest Classifier with 100 estimators.
- Evaluate the model on the validation set using accuracy score.
- Generate predictions on the test set and prepare a Kaggle submission file.

8. Results

The trained Random Forest model was evaluated on the held-out validation set (20% of the training data).

Metric	Value
Model Used	Random Forest Classifier (100 estimators)
Model Accuracy	82.12%
Train-Test Split	80% Train / 20% Validation
Random State	42

8.1 Feature Importance

The Random Forest model's built-in feature importance scores reveal which passenger attributes were most influential in predicting survival. Sex emerged as the strongest predictor, followed closely by Fare and Age, while Embarked had the least influence.

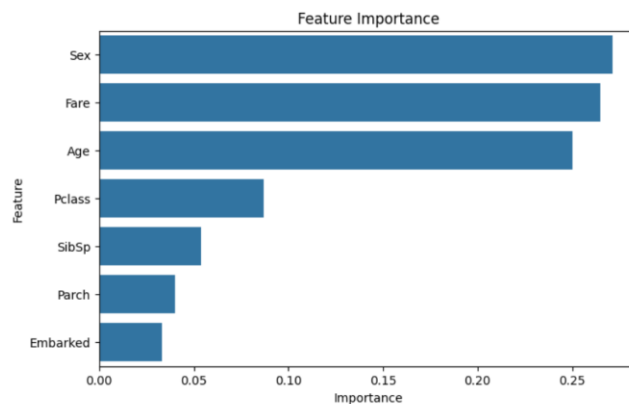


Figure 8.1 — Relative Importance of Each Feature in the Random Forest Model

This aligns closely with the historical record: gender played the single largest role in survival due to evacuation priority, while ticket fare (a proxy for socioeconomic status and cabin location) and age (children prioritized for lifeboats) were also strong determinants.

9. Conclusion

This project successfully demonstrates the complete machine learning workflow applied to the Titanic Survival Prediction problem — from raw, incomplete data to a working predictive model with measurable performance.

The Random Forest Classifier achieved an accuracy of 82.12% on the validation dataset, indicating that passenger attributes such as sex, class, age, and fare carry strong predictive signal regarding survival outcome.

Exploratory data analysis confirmed well-documented historical patterns: female passengers and those in higher ticket classes had substantially higher survival rates, consistent with the evacuation procedures followed during the disaster. The feature importance analysis further validated these findings quantitatively, with Sex, Fare, and Age identified as the three most influential predictors.

Overall, this project reinforced practical skills in data cleaning, exploratory analysis, feature engineering, and supervised classification modeling — forming a strong foundation for more advanced machine learning work.

10. Future Scope

- Apply advanced ensemble algorithms such as XGBoost, LightGBM, or Gradient Boosting to potentially improve predictive accuracy.
- Engineer additional features such as Title (extracted from passenger names), Family Size (SibSp + Parch), and Deck (extracted from Cabin) to capture more signal.
- Perform hyperparameter tuning using GridSearchCV or RandomizedSearchCV to optimize the Random Forest configuration.
- Apply cross-validation (e.g., k-fold) for a more robust estimate of model performance.
- Use SHAP or LIME for more detailed, instance-level model interpretability.
- Deploy the trained model as a web application using Flask or Streamlit for interactive predictions.